

Project Repositories Summary

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Project Repositories Summary

Workspace Inventory Report

/home/markt/proj

Generated from local repository inspection

Snapshot dates here (the **inspected on** line, maintenance section headings, and **Verification stamp**) use **this workstation's local calendar day** — i.e. what date reports under your current **TZ** / system clock — **not** UTC, unless a note says otherwise.

This directory contains 9 Git repositories. The notes below reflect each repo's local README.md, top-level files, and Git status as inspected on **2026-05-06**.

2026-05-06 maintenance update

- Re-surveyed all nine sub-repos for newest commits, branch alignment, and working-tree noise.
- **Verification stamp:** on **2026-05-06**, each clone matches **origin (ahead 0 / behind 0** after **git fetch**) and local dirt matches the **Repository notes** section (NOTES, generated UVM PDF) where called out.
- **chi-to-bow-bridge:** merged PR #19 — shared Pandoc defaults (**XeLaTeX** + **DejaVu** fonts), Makefiles use **--defaults** with **docs/pandoc-pdf-defaults.yaml** for design/integration/UVM/Verilator README PDFs; UVM Markdown reflow for PDF-safe layout (docs/pandoc-pdf.md). Repo remains Cocotb + Icarus core with **vlate_bench/** Verilator parity TB and optional **uvm_bench/** VCS/UVM.
- **IP-axi-to-2apbs:** documentation refresh — UVM README maps, PDF-friendly trees, **readme-md-pdfs** Make target.
- **IP-ucie-rdi-to-pcie-pipe:** UVM documentation PDF formatting polish; working tree still has **untracked** NOTES and test/uvm/UVM_README.pdf.
- **axi4_to_dfi_ddr:** added **UVM DV environment for VCS**; working tree has **untracked** NOTES.
- **snn-crossbar-model:** working tree has **untracked** NOTES (last tracked commit unchanged since roadmap add).
- **ucie-cxl-bridge:** cleanup of tracked tool artifacts and .gitignore refresh.

Prior sweep (**2026-05-02**) covered Verilator coverage on UCIE IP, AXI/DDR checkpoints, RISC-V layout refactor, SNN visualization/roadmap, CXL Phase 3 doc updates, AXI-2APBS Makefile wiring, and **pc-wsl-github-starter** CLI/tooling upgrades — details remain accurate where commits below did not move.

At a glance

- **IP-axi-to-2apbs:** AXI to dual-APB4 bridge RTL. Stack: Verilog, Icarus. State: main, clean.
- **IP-ucie-rdi-to-pcie-pipe:** UCIE RDI to PCIe PIPE bridge RTL. Stack: SystemVerilog, Verilator, vendor simulators. State: main, clean except untracked notes/PDF (see notes).
- **MIT_capstone_beetle_kill:** Forest bark beetle object detection. Stack: Python, PyTorch, pytest. State: main, clean.
- **axi4_to_dfi_ddr:** AXI4 to DFI / DDR bridge RTL. Stack: Verilog, Icarus, Verilator, Yosys, Pandoc, optional VCS/UVM DV. State: main, clean except untracked NOTES.
- **chi-to-bow-bridge:** CHI to BoW bridge starter. Stack: Verilog, Cocotb, Icarus, integration + **vlate_bench** Verilator TB, optional **uvm_bench** VCS/UVM, Pandoc/XeLaTeX PDFs. State: main, clean.
- **pc-wsl-github-starter:** WSL + GitHub starter workflow. Stack: Python, Typer, pytest, GitHub Actions. State: main, clean.
- **riscv_test_asm_qemu:** RISC-V cross-compile and QEMU experiments. Stack: RISC-V GNU toolchain, QEMU. State: master, clean.
- **snn-crossbar-model:** Spiking neural network crossbar model. Stack: Python, PyTorch, C++, SystemC, Verilog, pytest. State: main, clean except untracked NOTES.
- **ucie-cxl-bridge:** UCIE to CXL bridge RTL (Phases 1–3). Stack: Verilog/SystemVerilog, Icarus, Verilator, SymbiYosys. State: main, clean.

Repository notes

IP-axi-to-2apbs

- Purpose: AXI to 2× APB4 bridge RTL with self-checking testbenches.
- Highlights: separate simple and burst bridge variants, APB read wait-state test sweeps (make test-simple-ws-sweep), bridge design contract in doc/design_contract.md, UVM README maps / PDF-oriented trees, **readme-md-pdfs** Make target.
- Current state: clean working tree on main.
- Last commit: 2d985b6 — “docs: UVM README maps, PDF-friendly trees, readme-md-pdfs target”.

IP-ucie-rdi-to-pcie-pipe

- Purpose: production-style SystemVerilog bridge from UCIE 1.0 RDI to PCIe Gen4 PIPE.
- Highlights: dual-clock CDC with Gray-code pointer synchronization, per-lane elastic FIFOs, CRC32, CDC assertion monitors, self-checking scoreboard, Verilator coverage CI, FIFO stress verification plan, NUM_LANES=1 smoke test, reusable IP v1.0.0 baseline tag, enhanced UVM docs + PDF formatting.
- Notes: canonical RTL lives at the repo root; src/ and test/ subdirectories contain thin `include wrappers for EDA tools that prefer that layout. make verilator and make lint compile the root files directly.
- Current state: main aligned with origin/main; **untracked** NOTES and test/uvm/UVM_README.pdf present locally.
- Last commit: 213dd0e — “Enhance UVM documentation with professional PDF formatting”.

MiT_capstone_beetle_kill

- Purpose: PyTorch object-detection pipeline for aerial / oblique forest bark beetle survey imagery with Pascal-VOC-style XML annotations.
- Highlights: Faster R-CNN (ResNet50-FPN) training and inference, per-class precision/recall/F1 evaluation, CPU and GPU smoke tests, pytest suite, confidence-threshold tuning script, and dataset/process docs.
- Current state: clean working tree on main; PR #1 merged — automated test workflow, smoke assertion hardening, and agent metadata cleanup now on main.
- Last commit: f44c5fa — “Merge pull request #1: Improve detector robustness, test workflow, and agent metadata cleanup.”

axi4_to_dfi_ddr

- Purpose: AXI4 slave to JEDEC DFI-style bridge for a DDR PHY / memory-controller path.
- Highlights: gray-code async FIFOs for CDC, open-page SDRAM scheduler with per-bank PRE/ACT/CAS timing (MC_T_RP, MC_T_RCD, MC_T_RAS, MC_T_WR), optional refresh walk, AXI SLVERR error paths, elaboration-time parameter checks, Verilator lint, Yosys synthesis/formal hooks, generated PDF/HTML design spec, full-functionality roadmap (doc/FULL_FUNCTIONALITY_PLAN.md), **UVM DV environment for VCS**.
- Current state: main aligned with origin/main; **untracked** NOTES locally.
- Last commit: e671df3 — “feat(uvm_dv): add UVM DV environment for VCS”.

chi-to-bow-bridge

- Purpose: starter CHI-to-BoW bridge that packetizes simplified CHI requests into BoW flits and reconstructs responses.

- Highlights: Cocotb + Icarus unit/integration flows; integration top with reference BFM; OSS **vlate_bench/** Verilator + C++ parity TB (lint/run/coverage hooks); optional **uvm_bench/** Synopsys VCS / UVM bench with onboarding/quickref PDFs; integration protocol checker bind coverage across flows; BoW RX inject path + unknown-txn **RSP_HDR** parity; OSS regression Makefile targets; Markdown→PDF via Pandoc **XeLaTeX** + **DejaVu** (docs/pandoc-pdf-defaults.yaml, docs/pandoc-pdf-header.tex).
- Current state: clean working tree on main.
- Last commit: 00c0230 — “Merge pull request #19 from markrthomas/chore/pdf-pandoc-xelatex-dejavu”.

pc-wsl-github-starter

- Purpose: minimal starter repo for a Windows PC + WSL + GitHub workflow.
- Highlights: tiny Python app, unittest example, WSL-friendly Git config files, local make workflow, and a GitHub Actions CI.
- Current state: clean working tree on main.
- Last commit: 5283f74 — “Upgrade to Typer/Rich CLI, pytest, ruff/mypy, and venv Makefile.”

riscv_test_asm_qemu

- Purpose: RISC-V cross-compile and QEMU bring-up experiments covering core ISA fundamentals.
- Highlights: 10 test programs (hello, C++, exit-code, arithmetic, loops, function calls, memory load/store, bitwise ops, shifts, recursion); make test builds and verifies all 10 under QEMU with exact output and exit-code checks; source-interleaved disassembly targets; reorganized into src/asm/, src/cpp/, docs/, and build/ layout.
- Current state: clean working tree on master.
- Last commit: 7d878c7 — “refactor: reorganize into src/asm, src/cpp, docs, and build/.”

snn-crossbar-model

- Purpose: spiking neural network targeting hardware-oriented crossbar (RRAM/PCM) design, with four-way fixed-point cross-check across Python, C++, SystemC, and Verilog RTL.
- Highlights: snntorch-based training with noise-aware QAT, weight-level × timestep × hidden-dim sweep, Gaussian noise robustness evaluation, bit-identical four-way cross-check (Python == C++ == SystemC == Verilog), synthesizable Verilog core (snn_core_fixed.v) with multi-cycle FSM and start/done/busy handshake, visualization tool (scripts/visualize.py), and 38-test pytest suite.
- Current state: main aligned with origin/main; **untracked** NOTES locally.
- Last commit: cb14e44 — “Add development roadmap.”

ucie-cxl-bridge

- Purpose: experimental UCIE Adapter Layer to CXL.io / CXL.cache / CXL.mem bridge.
- Highlights: three completed phases — Phase 1 typed 64-bit packet model and CXL.io <-> UCIE translation; Phase 2 full packet taxonomy (CXL.io / CXL.mem / CXL.cache requests and AD_CPL / MEM_CPL / CACHE_CPL completions) with XOR checksum and GTKWave save file; Phase 3 per-direction credit counters (credit_counter.v), posted/non-posted ordering domain split with posted-priority egress arbiter and registered arbiter lock for valid/ready stability, ordering directed test, and SymbiYosys BMC + cover on both sync_fifo and cxl_ucie_bridge. Verilator lint clean, Windows PowerShell sim helper included.
- Current state: clean working tree on main.
- Last commit: 12b04a6 — “Clean up tracked tool artifacts and update .gitignore”.

Overall observations

- Eight of the nine repositories are hardware / verification projects (RTL bridges, CDC, memory controllers, a protocol bridge, and a spiking neural network with RTL cross-check).
- `MiT_capstone_beetle_kill` is the sole ML/Python project.
- All nine repositories track **main** / **master** in lockstep with their remotes; several clones carry **untracked** scratch (NOTES) or generated UVM PDFs as noted above.
- **Workspace PDF**: this summary's **make pdf** uses the same **XeLaTeX + DeJaVu** stack as **chi-to-bow-bridge** (`.pandoc-pdf-defaults.yaml` + `.pandoc-header.tex`).